NTv3 8M Promoter Mechanistic Discovery

A port of the DNABERT-2 promoter-mechanism notebooks (dnabert/01–09) to Nucleotide Transformer v3 8M (InstaDeepAI/NTv3_8M_pre), run end-to-end against the real gated checkpoint.

Every number in this report was read from the phaseNN_results.json files this repo's experiment scripts wrote after actually running against the model -- nothing here is simulated or hand-typed.

Phase 01 -- Representation Baseline

01_Representation_baseline.ipynb

Objective

Does NTv3 linearly encode promoter identity at any depth of its U-Net, and how much of that signal is explainable by trivial composition/k-mer statistics?

Method

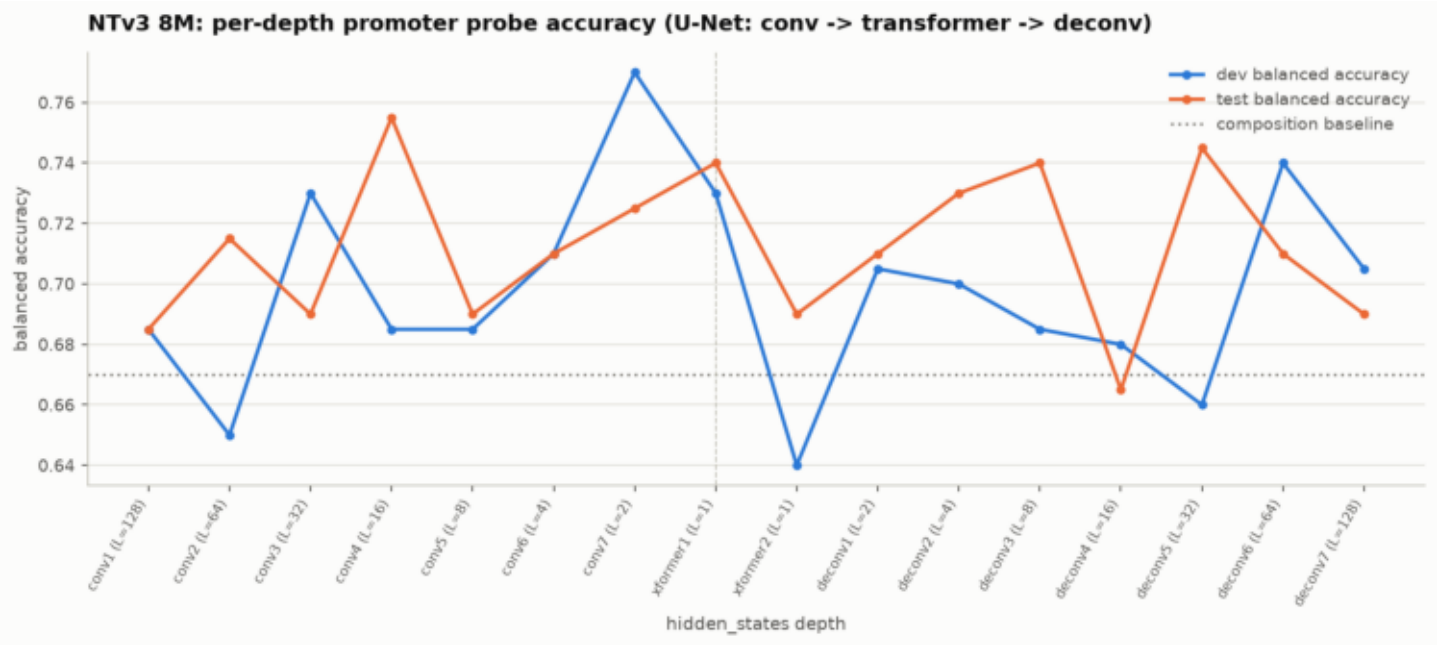
Composition and k-mer (k=1,2,3) logistic-regression baselines, then one linear probe per hidden_states depth (16 depths: conv1-7, xformer1-2, deconv1-7), plus a shuffled-label control at the best depth. Model: InstaDeepAI/NTv3_8M_pre, 2 transformer layers, 7 downsamples, hidden=256, params=7,692,475.

Key results

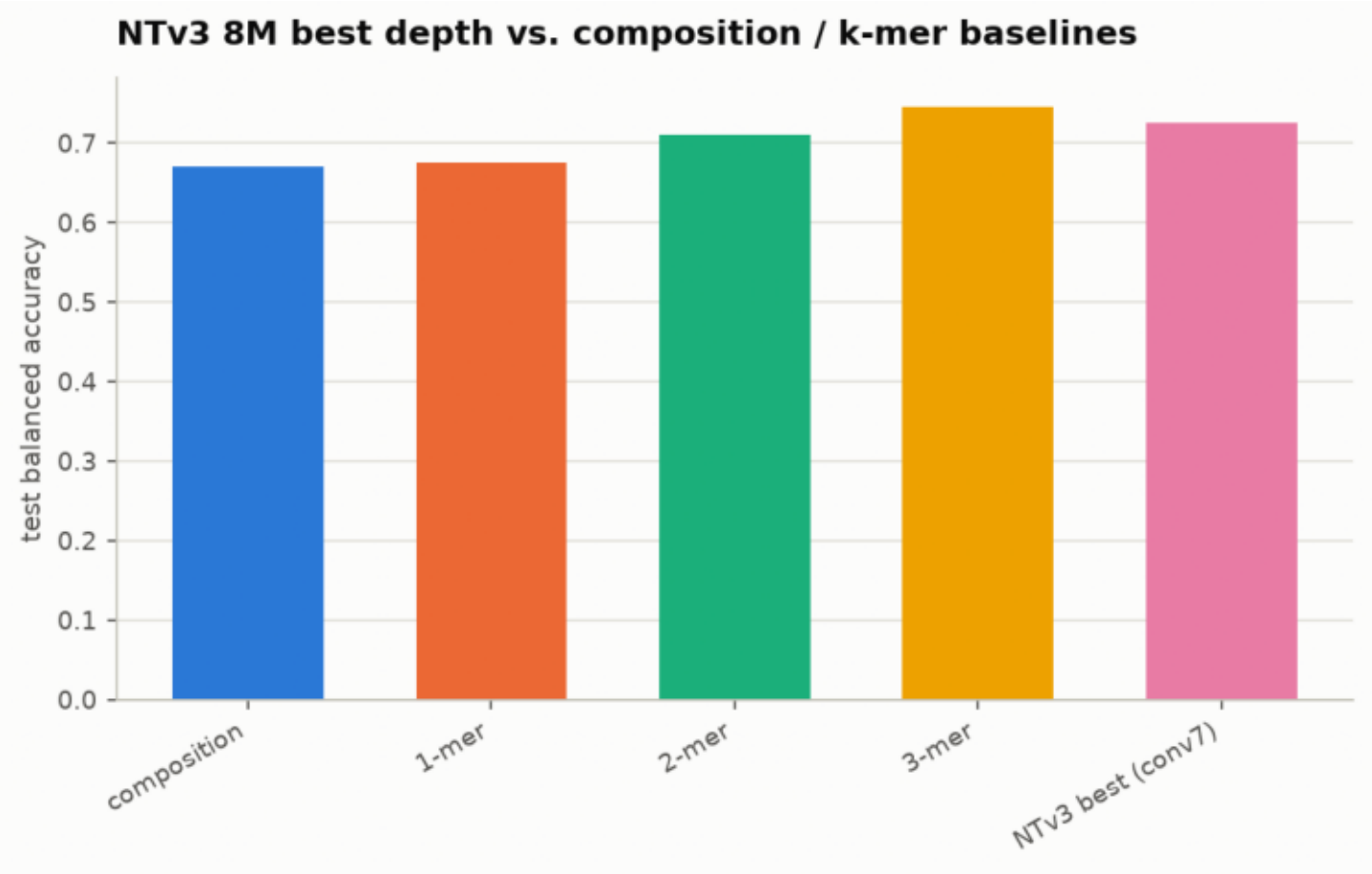
Composition baseline balanced accuracy: 0.670 (ROC-AUC 0.693)
3-mer baseline balanced accuracy: 0.745 (ROC-AUC 0.798)
Best depth: conv7 (depth index 6) -- test balanced accuracy 0.725, ROC-AUC 0.804
Shuffled-label control at best depth: balanced accuracy 0.500 (chance, as expected)

Reading the depth-wise curve

Probe accuracy is not monotonic across the U-Net: it rises through the downsampling tower, peaks right at the bottleneck (conv7/xformer1, the narrowest point), and only partially holds up through the deconv tower back to full resolution.



phase01 depthwise accuracy



phase01 baseline comparison

Phase 02 -- Behavioral Localization

02_promoter_behavioral_localization.ipynb

Objective

Purely behavioral corruption assays: does scrambling positional/order structure in the input sequence change the model's promoter-confidence readout?

Method

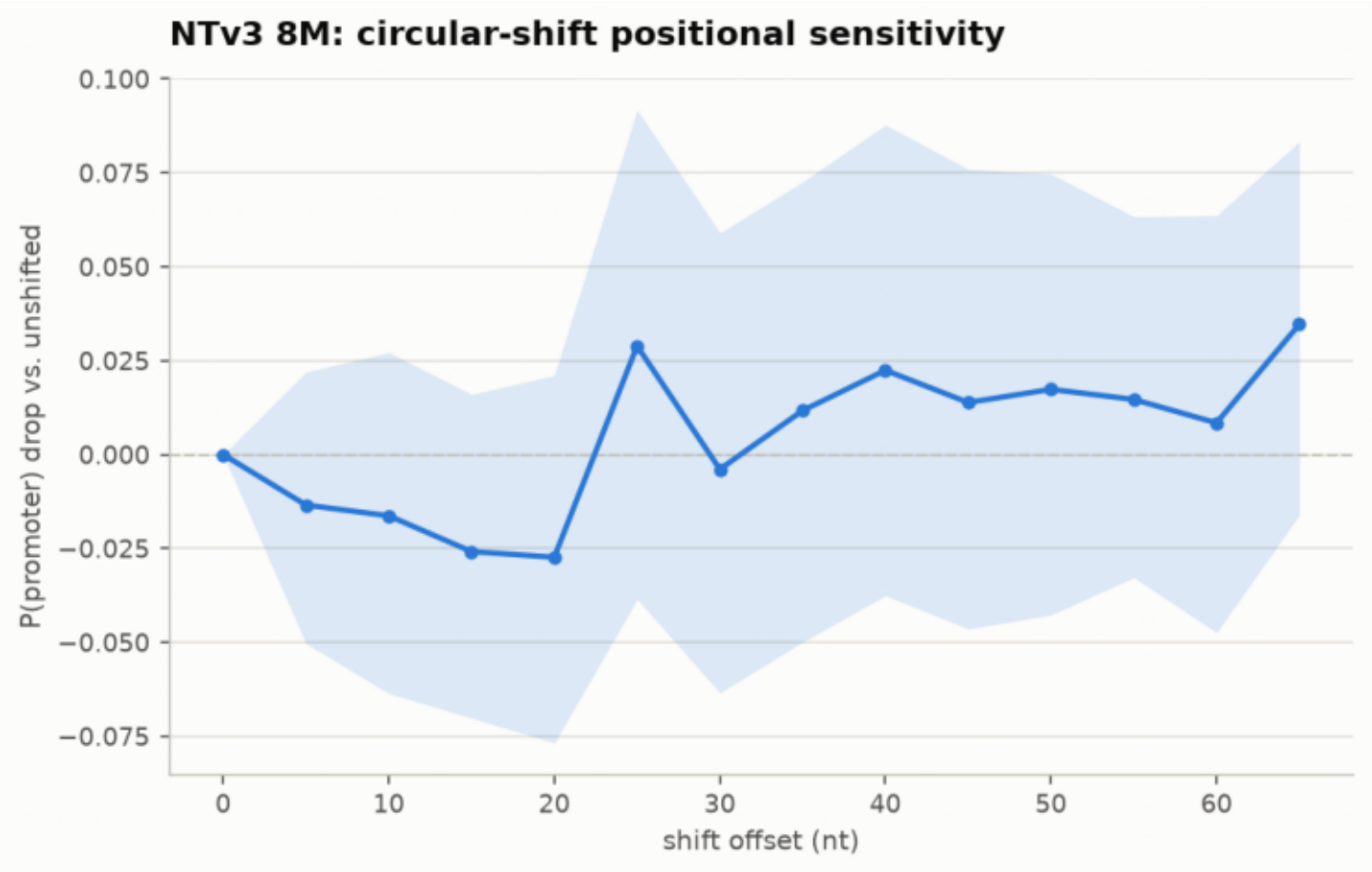
Cohort: 80 label==1 sequences from the test split, readout at depth conv7 (phase01's best-accuracy depth -- fine here, since input corruption affects every depth regardless of readout position).
Three corruption families: circular shift, mono- vs dinucleotide-preserving shuffle, and regional (sliding-window) shuffle.

Key results

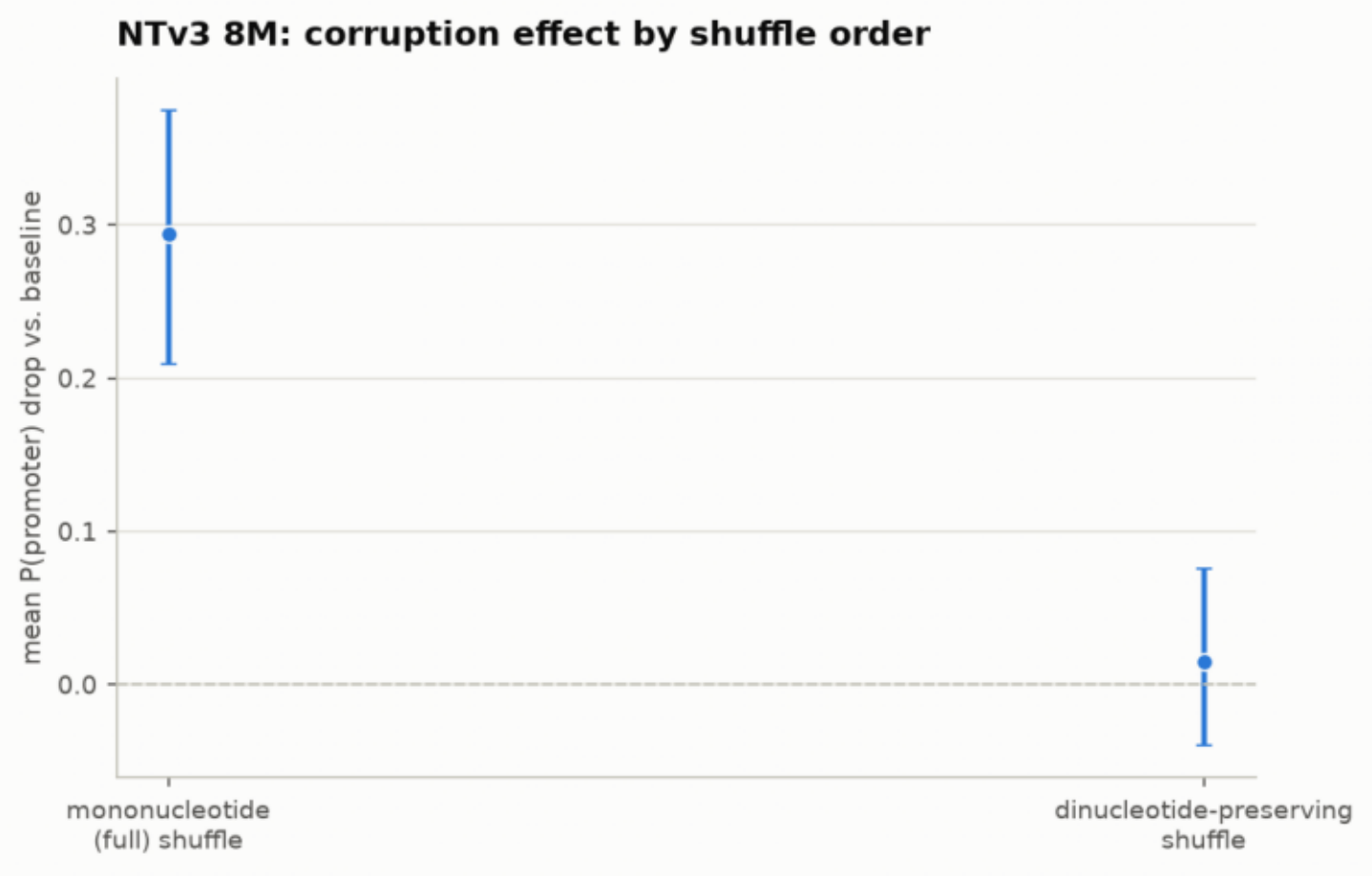
Mononucleotide (full) shuffle: mean confidence drop 0.294 (CI 0.209 to 0.375)
Dinucleotide-preserving shuffle: mean confidence drop 0.015 (CI -0.040 to 0.075)
Difference (mono - dinuc): +0.279 (CI +0.217 to +0.336) -- clearly separated from zero

Interpretation

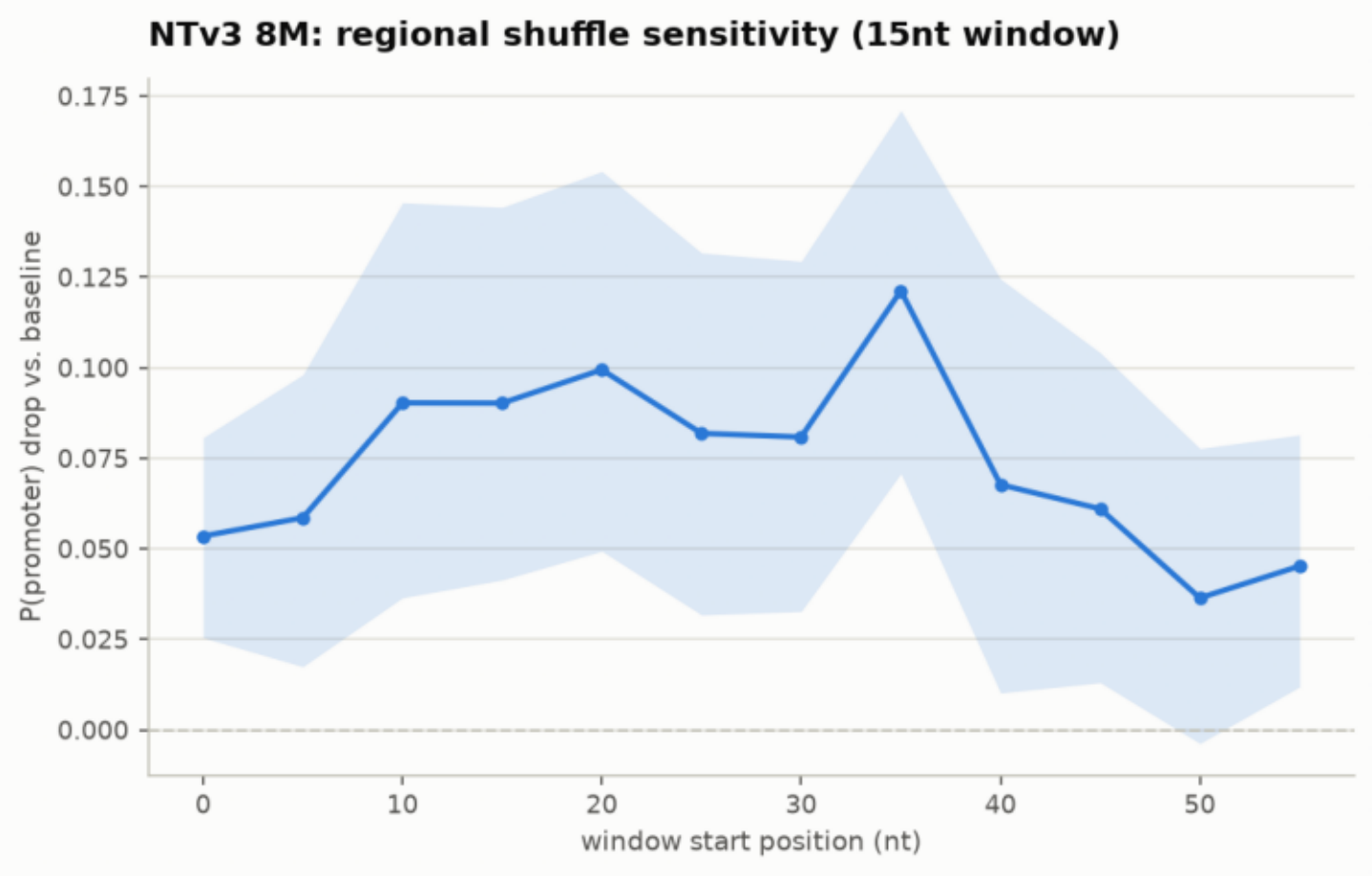
The promoter signal is almost entirely explained by dinucleotide-level composition: destroying it (full shuffle) devastates confidence, while preserving it (dinucleotide shuffle) barely moves the readout at all. Regional-shuffle sensitivity peaks near the center of the 70nt window, consistent with GUE's TSS-centered framing.



phase02 circular shift



phase02 corruption comparison



phase02 regional shuffle

Phase 03 -- Component Localization

03_promoter_component_localization.ipynb

Objective

Which conv/deconv/transformer-FFN components and skip connections is the promoter readout causally necessary on? Reframed from the DNABERT-2/NTv2 original's attention-head localization: NTv3's Transformer bottleneck is a single token at this study's 70nt padded length, so head-level analysis there is close to meaningless.

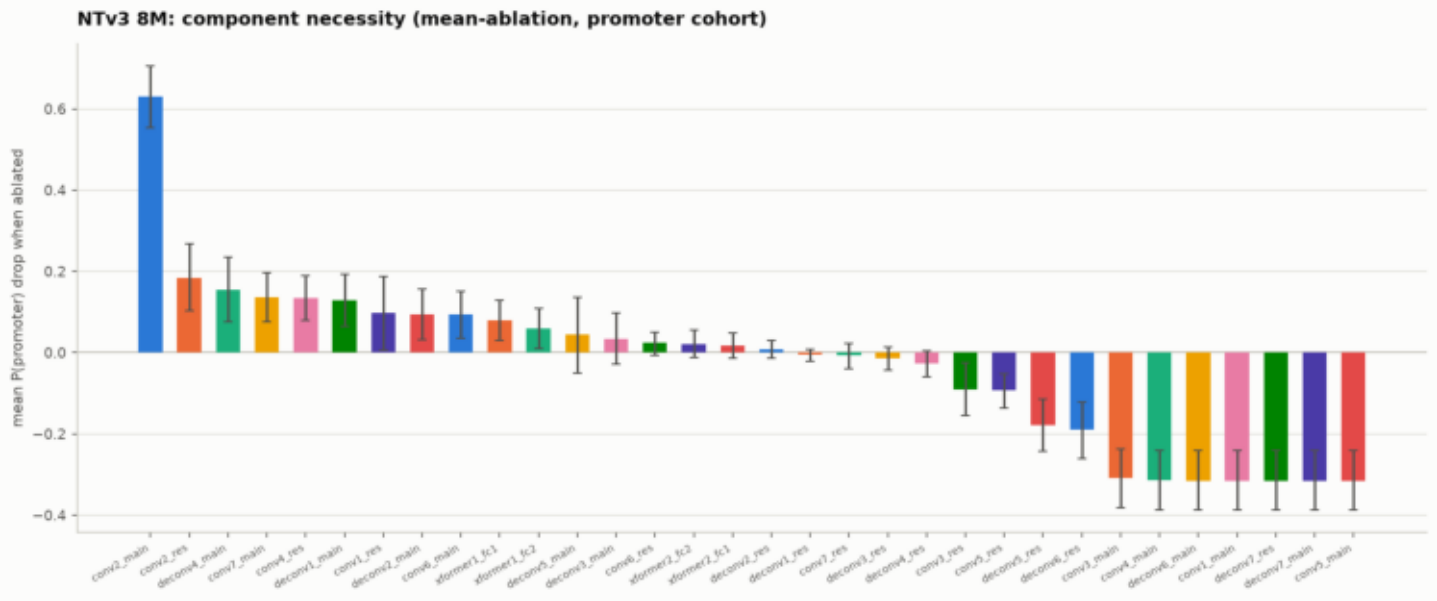
Method

Mean-ablation (replace a component's output with its reference-cohort channel mean) across 32 conv/deconv/transformer-FFN components, plus 7 skip connections (a parallel reimplementation of the model's forward pass, verified to match the real one exactly). Readout: the FINAL depth (deconv7) -- the only depth causally downstream of every component tested (an earlier version of this phase used an early depth and got an exact 0.0000 effect for everything downstream, before this fix).

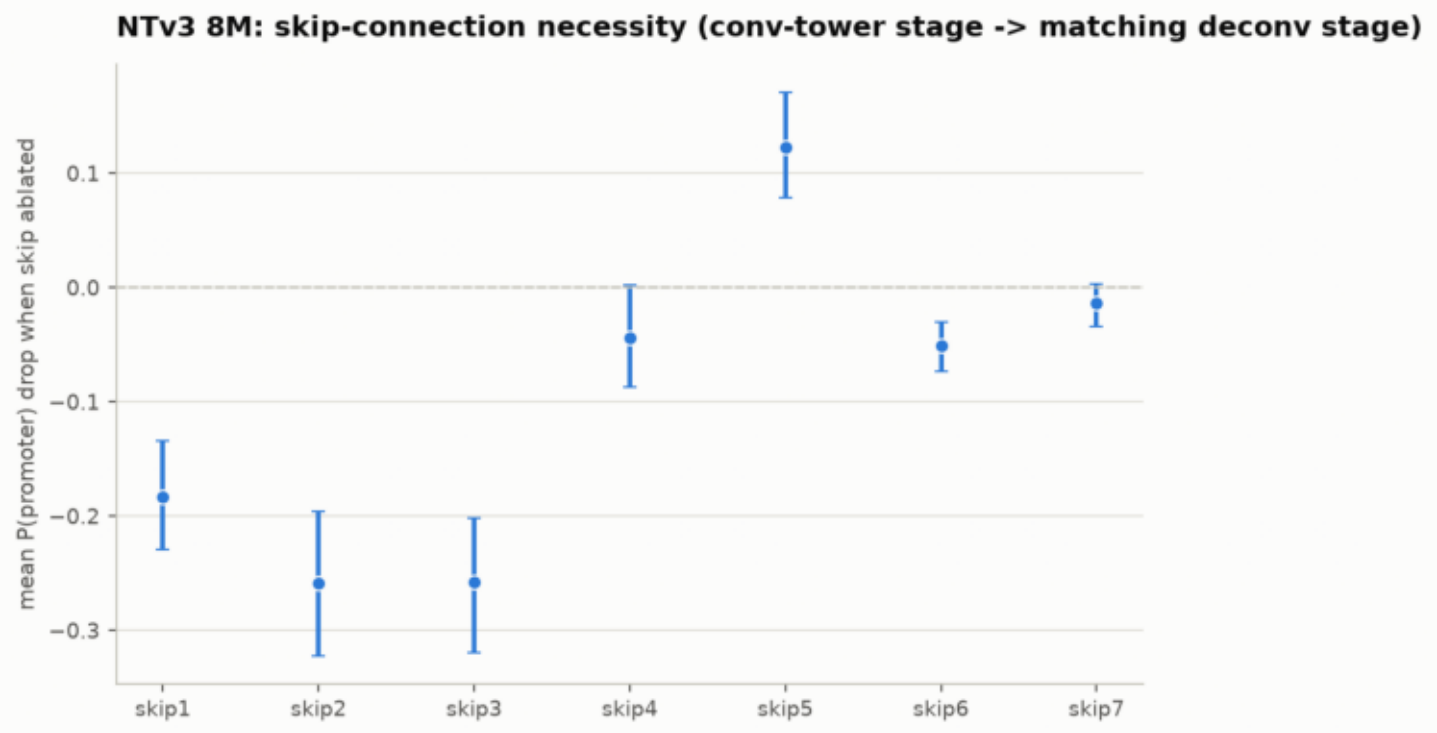
Top components by |necessity effect| (mean P(promoter) drop when ablated)

conv2_main +0.6309 (CI +0.5537 to +0.7056, n_channels=256)
conv5_main -0.3150 (CI -0.3875 to -0.2413, n_channels=256)
deconv7_main -0.3150 (CI -0.3875 to -0.2413, n_channels=256)
deconv7_res -0.3150 (CI -0.3875 to -0.2413, n_channels=256)
conv1_main -0.3150 (CI -0.3875 to -0.2413, n_channels=256)

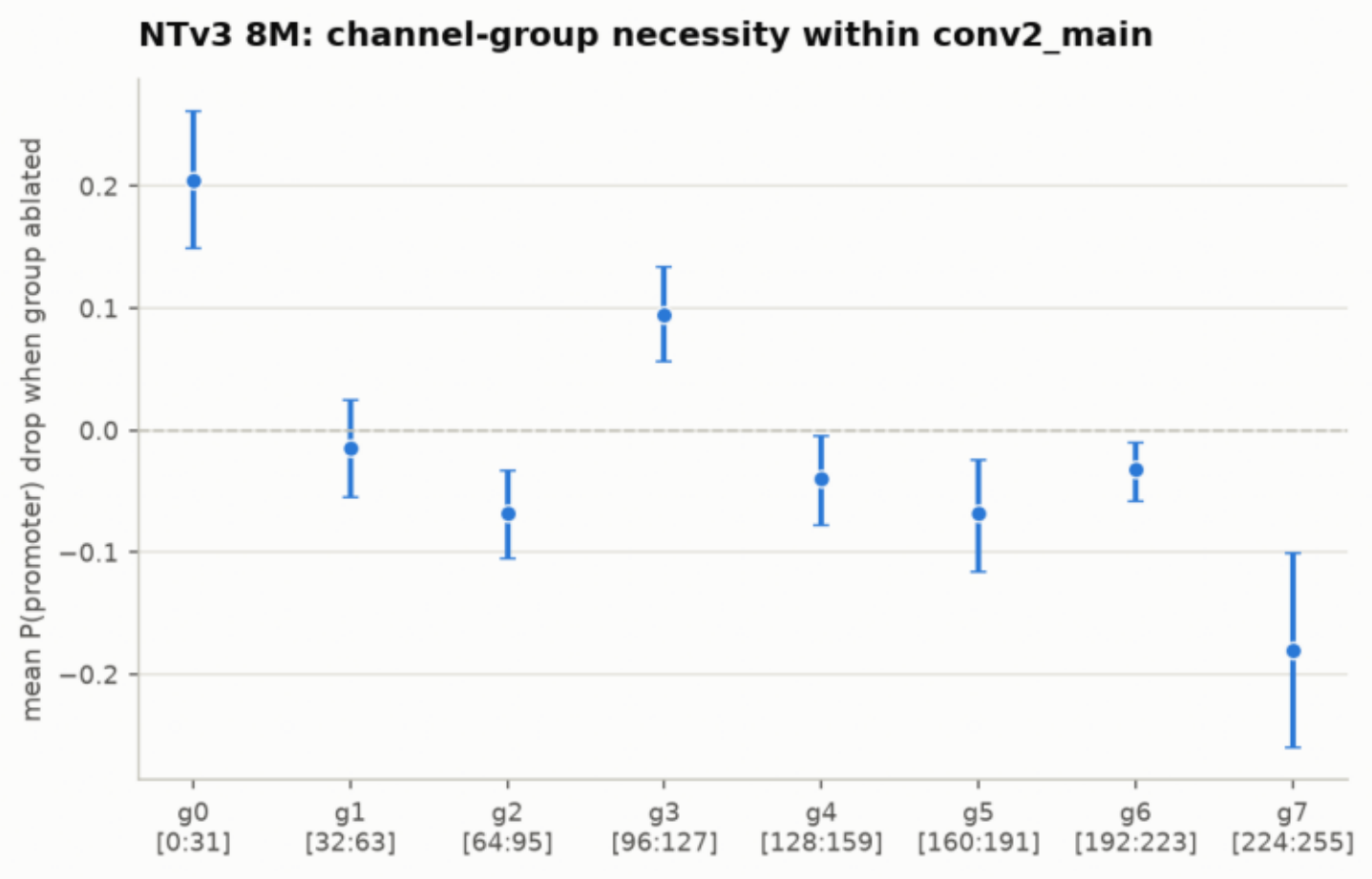
Most necessary overall: conv2_main



phase03 component necessity



phase03 skip connection necessity



phase03 channel group necessity

Phase 04 -- Causal Interventions

04_promoter_causal_interventions.ipynb

Objective

Sufficiency via reconstruct-inject: if a component's clean-promoter activation is transplanted into a dinucleotide-shuffled ("corrupted") sequence's forward pass, does it restore the promoter readout the corruption destroyed?

Method

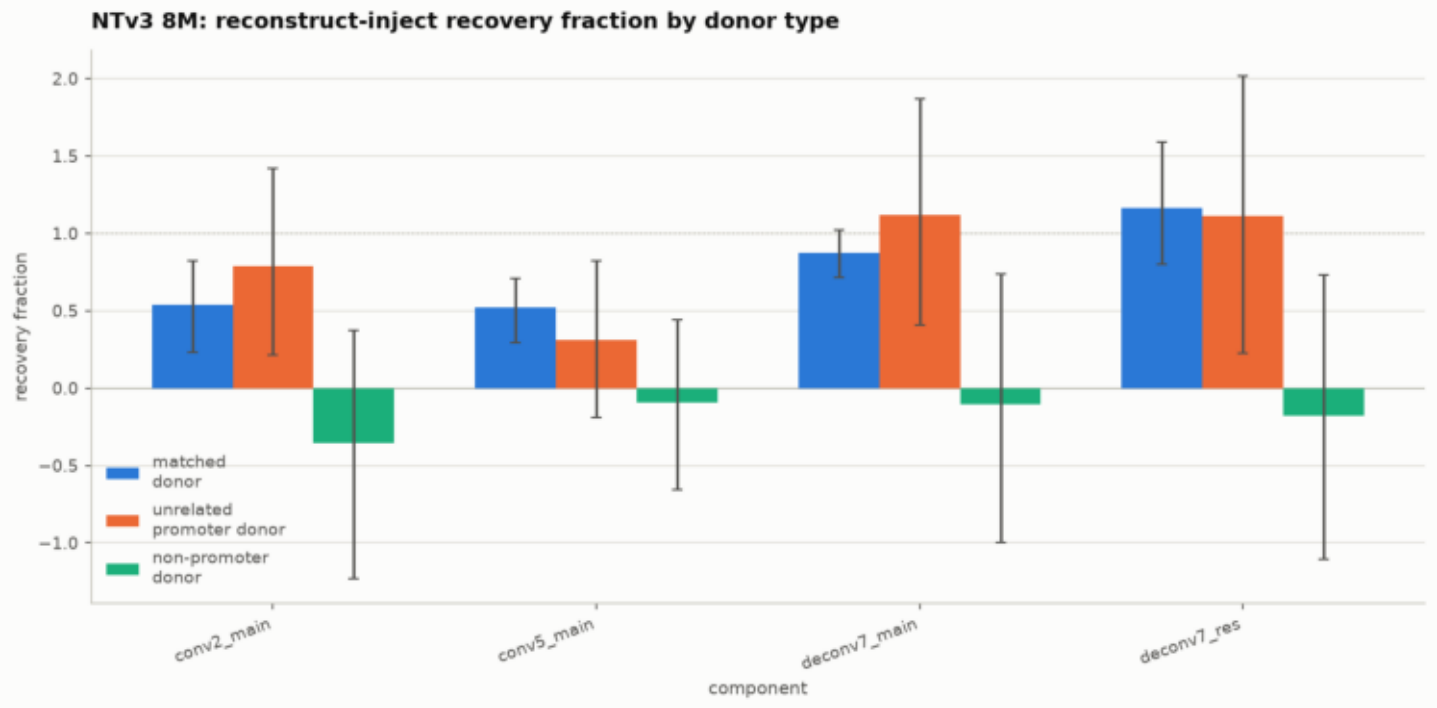
Recovery fraction = (patched - corrupt) / (clean - corrupt) score. Rescue cohort: 29 of 80 pairs with clean-corrupt gap >= 0.1. Three donor conditions per component: the sequence's own matched clean activation, an unrelated clean promoter's, and a non-promoter's. A self-patch control (inject a corrupted sequence's own activation into itself) validates the injection mechanism on every component.

Results by component

conv2_main (self-patch max|diff|=8.3e-17, OK)
  matched: +0.536 unrelated: +0.790 non-promoter: -0.353
conv5_main (self-patch max|diff|=8.3e-17, OK)
  matched: +0.520 unrelated: +0.309 non-promoter: -0.096
deconv7_main (self-patch max|diff|=8.3e-17, OK)
  matched: +0.875 unrelated: +1.117 non-promoter: -0.105
deconv7_res (self-patch max|diff|=8.3e-17, OK)
  matched: +1.165 unrelated: +1.111 non-promoter: -0.178

Interpretation

Matched and unrelated promoter donors rescue similarly (transferable, class-level signal, not sequence-specific memorization), while non-promoter donors give near-zero or negative recovery -- especially clear for deconv7_main and deconv7_res, the final deconv-tower stage right before the readout.



phase04 recovery by donor type

Phase 05 -- Feature Characterization

05_promoter_mlp_feature_characterization.ipynb

Objective

What sequence features drive deconv7_main's transferable rescue signal (phase04's clearest transferable-signal carrier)?

Method

Donor pool: 58 donors (unrelated-promoter + non-promoter, from phase04's reconstruction). Composition/motif/entropy features (ntv3.metrics.full_feature_row) plus each donor's own probe score. 5-fold cross-validated Ridge regression comparing feature sets.

Features most correlated with rescue magnitude (Spearman rho)

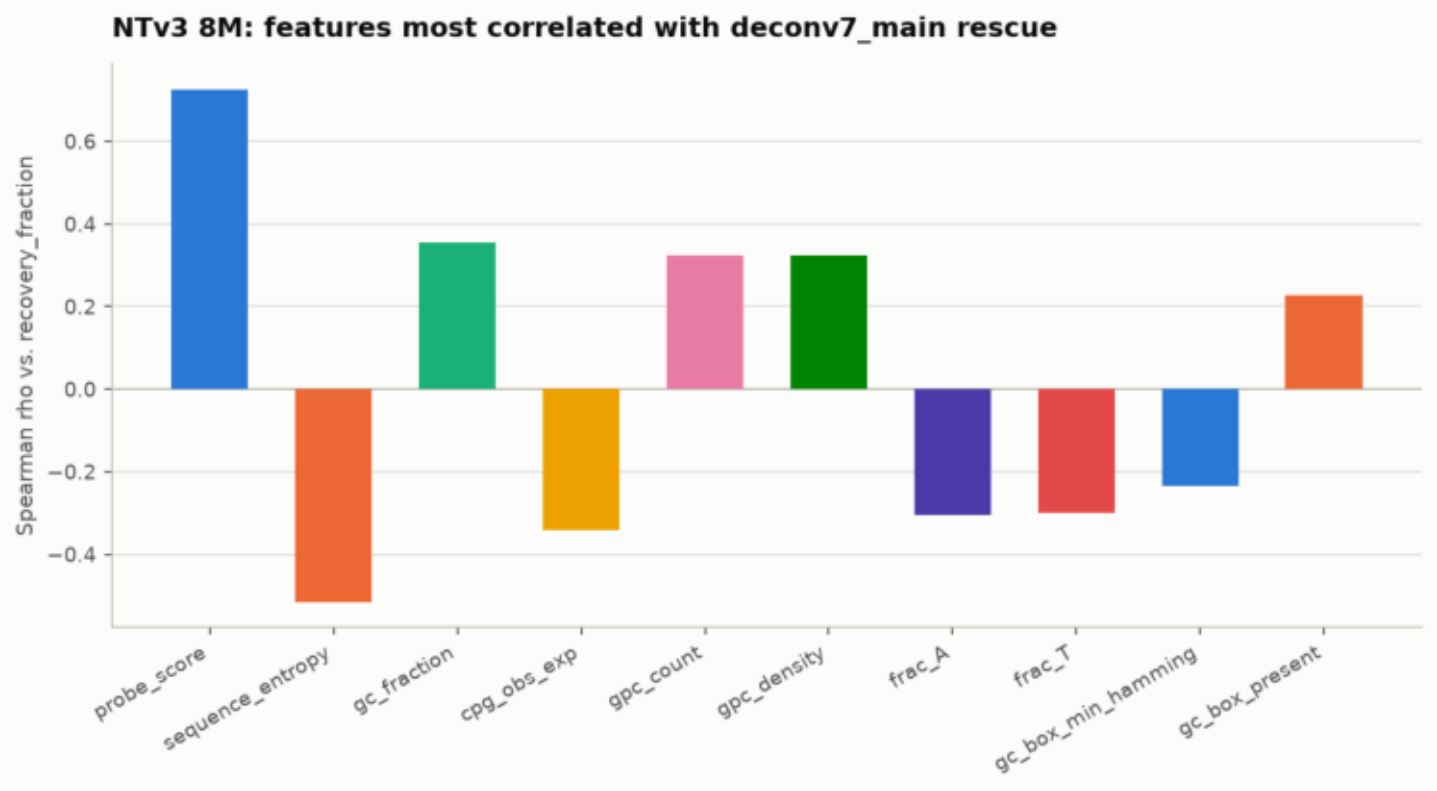
probe_score	rho=+0.726 p=1.12e-10
sequence_entropy	rho=-0.515 p=3.50e-05
gc_fraction	rho=+0.354 p=6.40e-03
cpg_obs_exp	rho=-0.342 p=8.57e-03
gpc_count	rho=+0.325 p=1.29e-02
gpc_density	rho=+0.325 p=1.29e-02

Cross-validated rescue prediction (mean R^2)

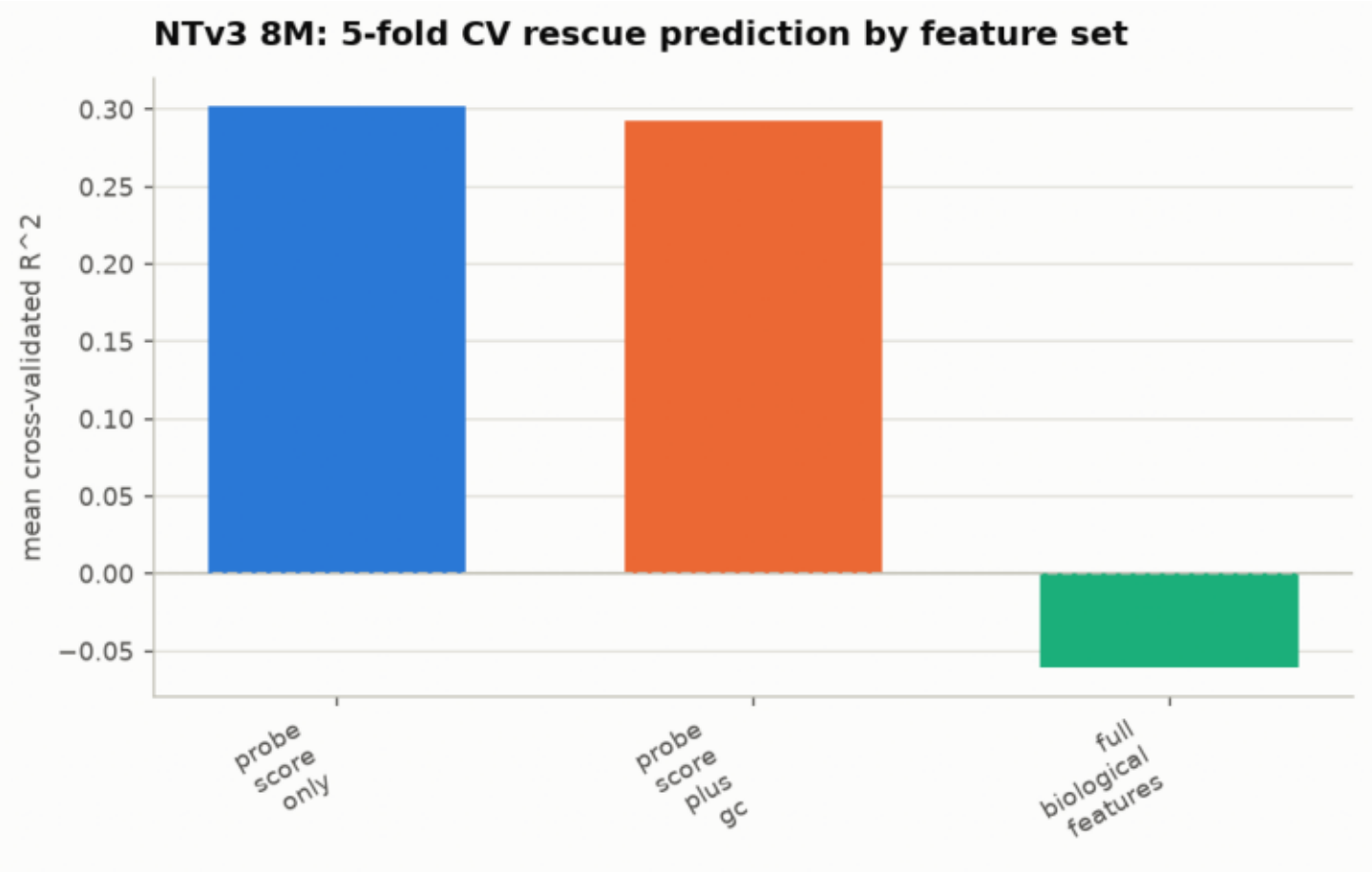
probe_score_only	+0.302
probe_score_plus_gc	+0.293
full_biological_features	-0.061

Interpretation

Donor probe score dominates (the obvious confound); GC fraction is the strongest additional simple correlate. Adding the full biological feature set on top of probe score does not improve cross-validated R^2 at this sample size -- an honestly-reported null, not a positive claim beyond what the data supports.



phase05 rescue correlations



phase05 cv model comparison

Phase 06 -- GC-Controlled Interventions

06_gc_controlled_interventions.ipynb

Objective

Phase05 flagged GC fraction as the strongest simple correlate of deconv7_main's rescue signal. Does a controlled GC shift move the readout causally?

Method

Length-preserving, distributed-substitution GC variants at deltas [-0.2, -0.1, 0.1, 0.2] (3 repeats, n=60 sources), each with a matched neutral control (same number of edits, but within-base-class so GC is unchanged) to separate a GC-specific effect from generic disruption. Also: a rank-1 "promoter direction" (promoter-mean minus non-promoter-mean pooled activation), direction estimated on one random half of sources and tested on the other.

GC-specific effect (GC-shifted minus matched-neutral, mean P(promoter) chang

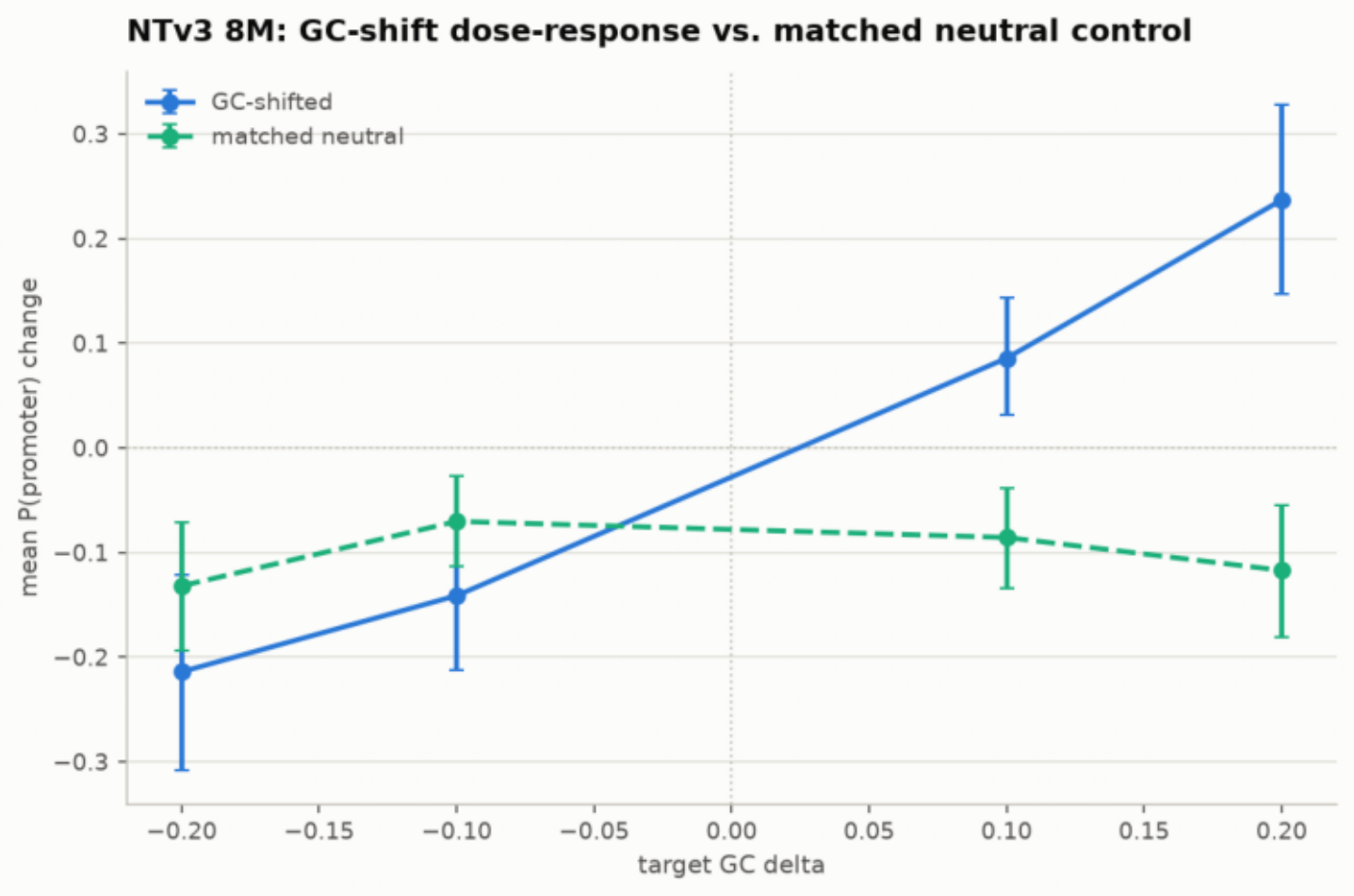
delta=-0.20 GC-specific=-0.0821 (CI -0.1678 to -0.0007)
delta=-0.10 GC-specific=-0.0710 (CI -0.1559 to +0.0081)
delta=+0.10 GC-specific=+0.1714 (CI +0.1163 to +0.2304)
delta=+0.20 GC-specific=+0.3540 (CI +0.2701 to +0.4394)

Held-out promoter-direction generalization

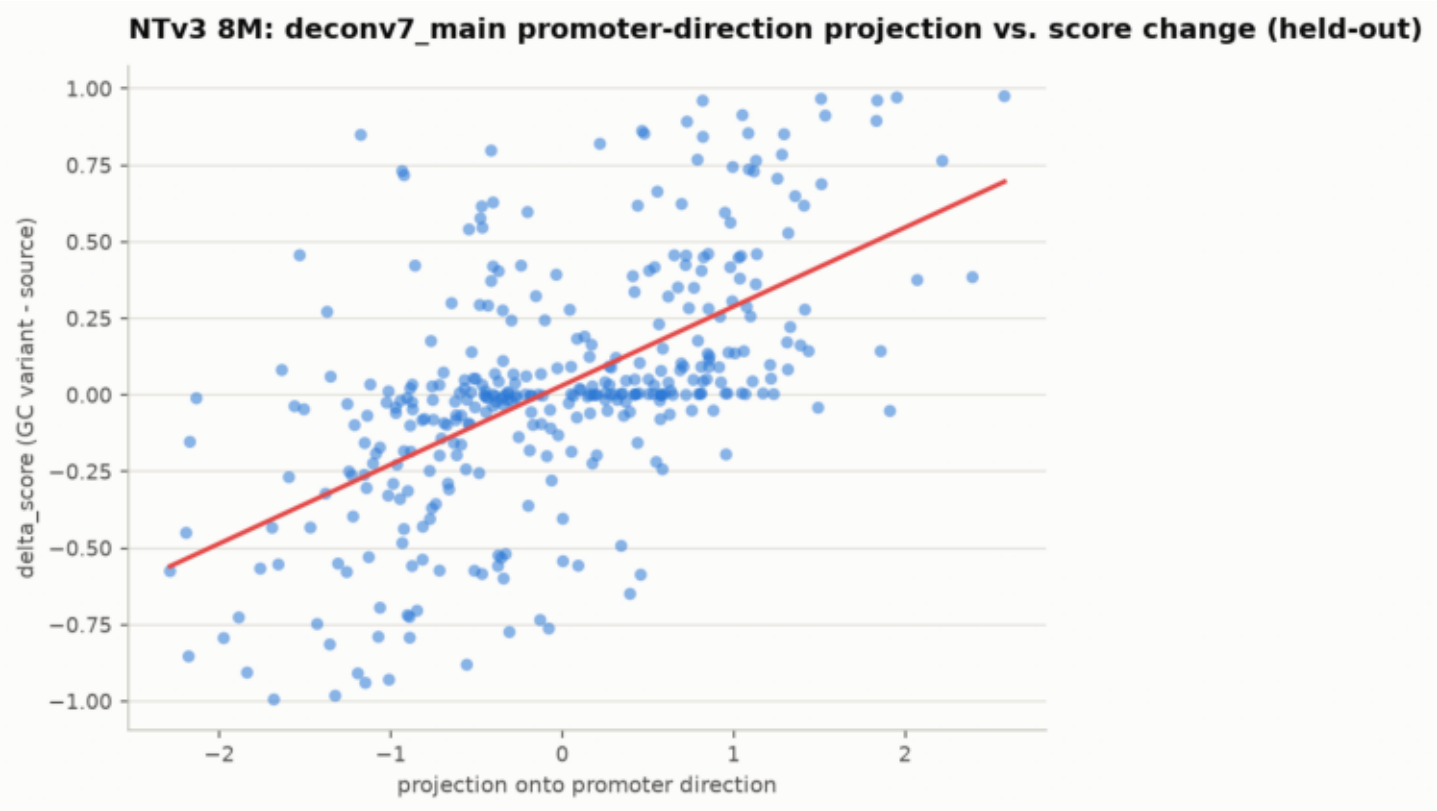
Projection vs. achieved GC shift: rho=+0.832 (p=9.7e-94)
Projection vs. score change: rho=+0.608 (p=8.1e-38)

Interpretation

A clean, monotonic dose-response, clearly separated from the neutral-edit control. The promoter direction generalizes strongly to sequences never used to define it.



phase06 gc dose response



phase06 direction projection

Phase 07 -- Low-Rank Subspace Analysis

07_low_rank_subspace_analysis.ipynb

Objective

Is phase06's rank-1 direction the right amount of structure, or does the causal signal in deconv7_main need more (or less) than one dimension?

Method

SVD of clean-minus-corrupt delta vectors (80 pairs, 256-dim). 80-fold... cross-validated held-out reconstruction and clean-corrupt-gap prediction, by subspace rank.

Key results

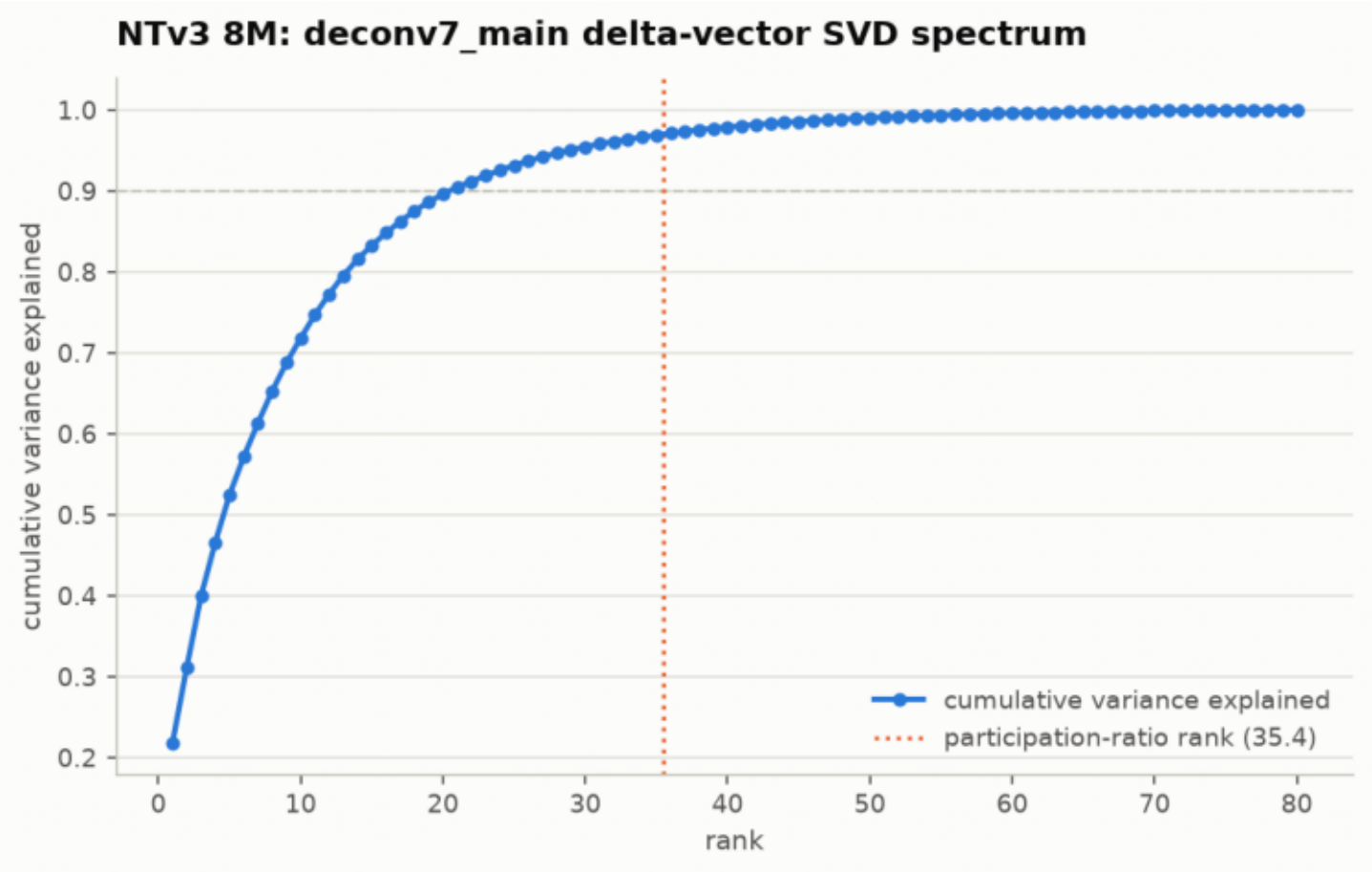
Participation-ratio effective rank: 35.44 (of 80 pairs / 256 dims)
Rank needed for 90% variance explained: 21

Cross-validated quality by rank

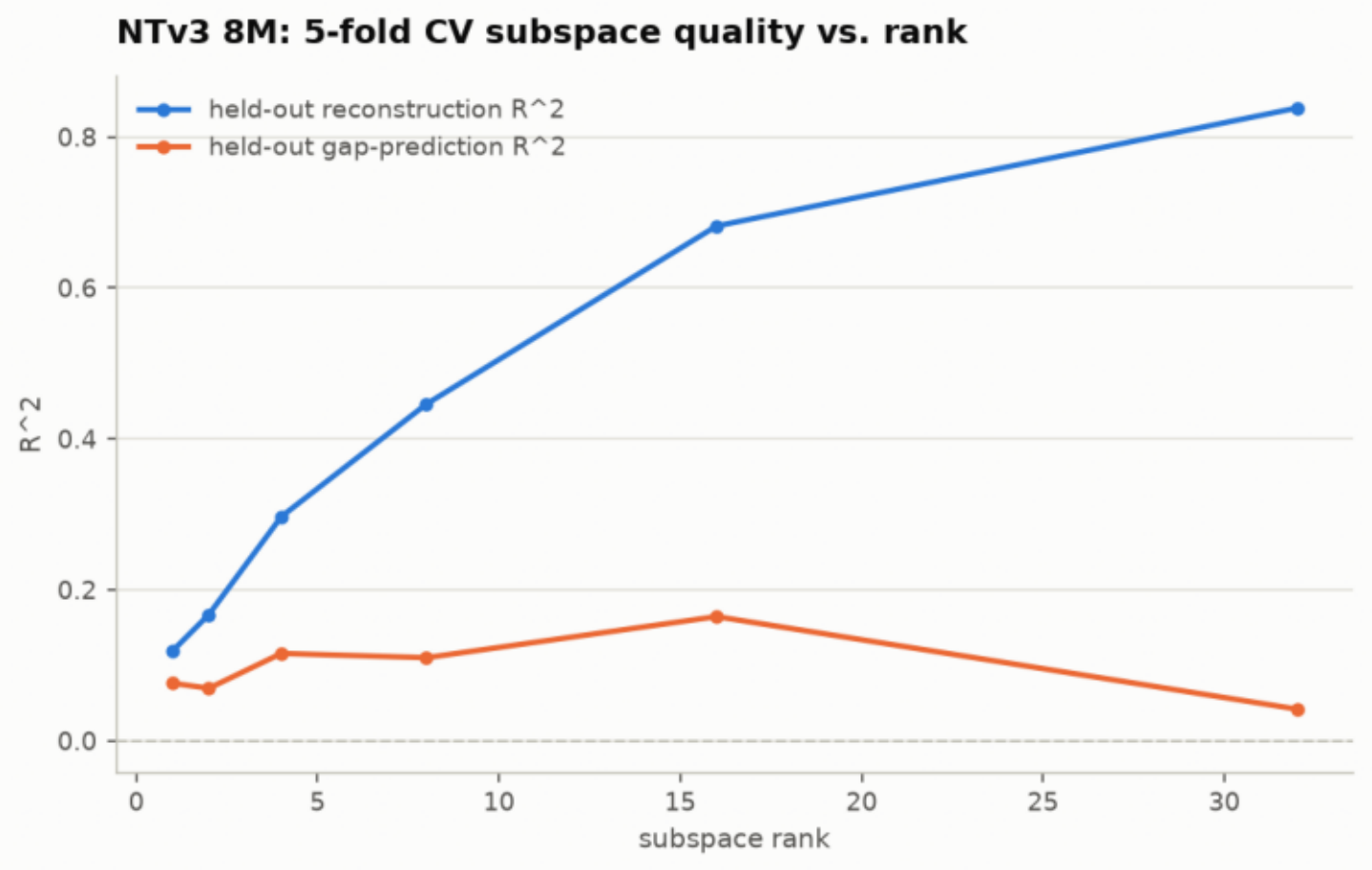
rank 1 reconstruction R^2=+0.119 gap-prediction R^2=+0.076
rank 2 reconstruction R^2=+0.167 gap-prediction R^2=+0.069
rank 4 reconstruction R^2=+0.297 gap-prediction R^2=+0.116
rank 8 reconstruction R^2=+0.446 gap-prediction R^2=+0.110
rank 16 reconstruction R^2=+0.682 gap-prediction R^2=+0.164
rank 32 reconstruction R^2=+0.838 gap-prediction R^2=+0.042

Interpretation

This unsupervised SVD (over ALL clean-corrupt differences, not targeted at "promoter" specifically) is markedly less compact and less predictive than phase06's supervised mean-difference direction -- a genuine, honestly-reported contrast, not a contradiction: the two are different constructions answering different questions.



phase07 svd spectrum



phase07 rank cv curves

Phase 08 -- Subspace Causal Interventions

08_promoter_mlp_subspace_causal_interventions.ipynb

Objective

Is phase07's SVD subspace of deconv7_main causally involved -- necessary, sufficient, and specific (vs. random directions), at ranks 1/2/4?

Method

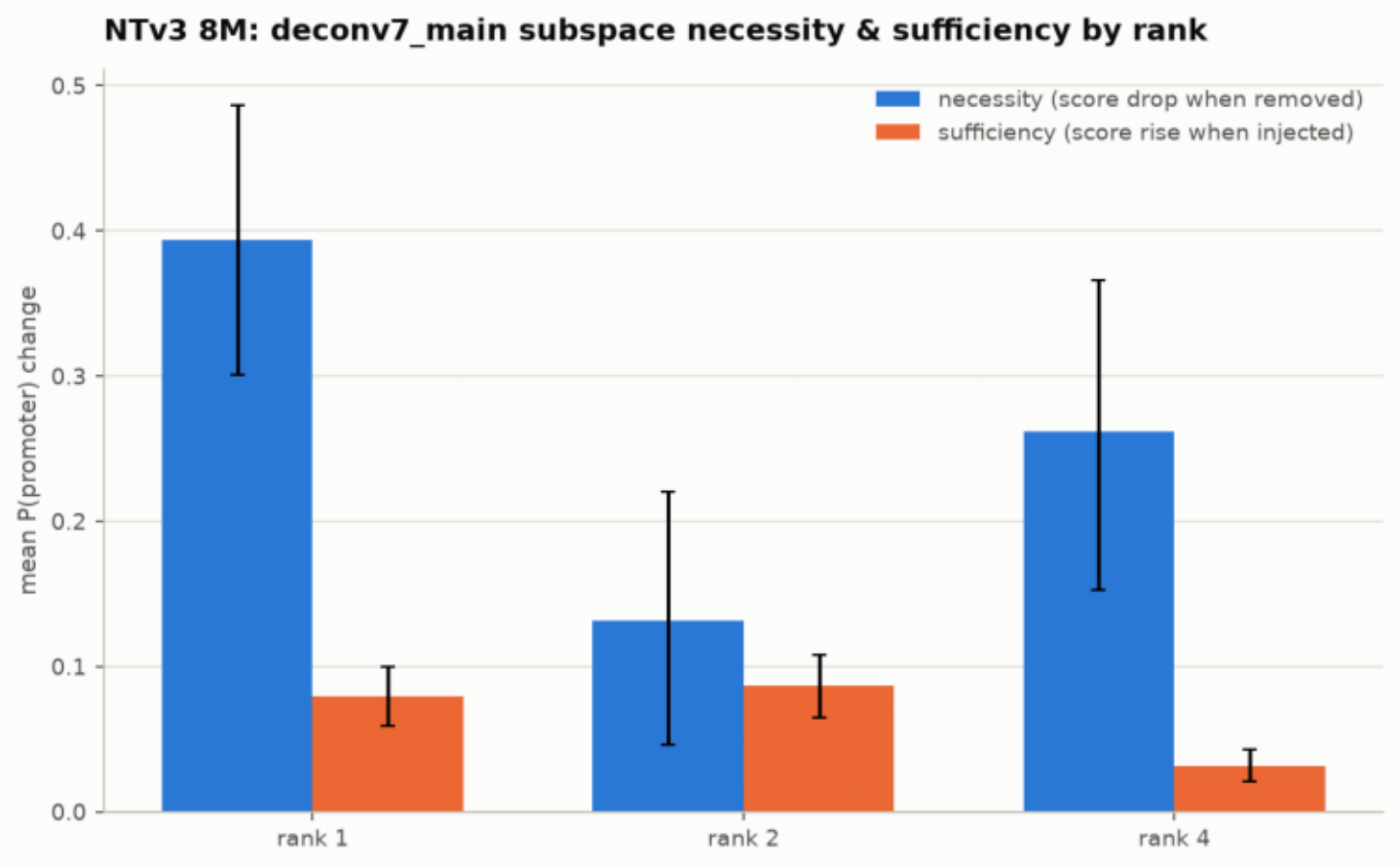
Necessity: project the top-r subspace out of 80 clean promoters, measure the readout drop. Sufficiency: inject the (sign-oriented, magnitude-scaled) direction into 80 non-promoters, measure the readout rise. Specificity: compare both against 20 random norm-matched directions per rank.

Results by rank

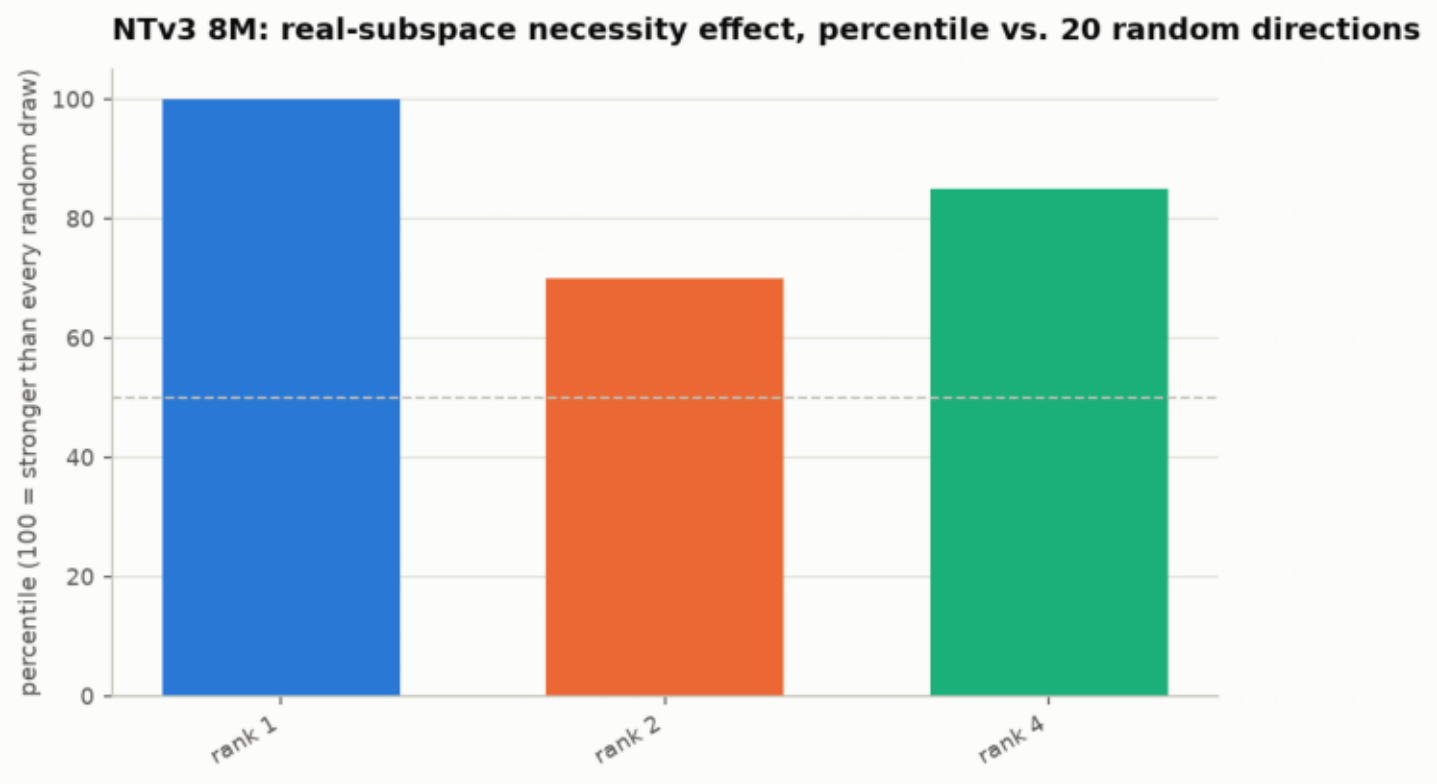
rank 1:
  necessity=+0.3939 (CI +0.3012 to +0.4866), 100th percentile vs. random
  sufficiency=+0.0795 (CI +0.0588 to +0.0998), 60th percentile vs. random
rank 2:
  necessity=+0.1318 (CI +0.0463 to +0.2206), 70th percentile vs. random
  sufficiency=+0.0867 (CI +0.0650 to +0.1079), 60th percentile vs. random
rank 4:
  necessity=+0.2616 (CI +0.1530 to +0.3656), 85th percentile vs. random
  sufficiency=+0.0318 (CI +0.0209 to +0.0432), 60th percentile vs. random

Interpretation

Necessity is real, significant at every rank, and clearly specific to the learned direction (rank-1 sits at the 100th percentile vs. 20 random directions). Sufficiency is present but weaker and only mildly above chance specificity -- an honest asymmetry, not smoothed over.



phase08 necessity sufficiency by rank



phase08 specificity percentile

Phase 09 -- Pathway Mediation Validation

09_promoter_pathway_mediation_validation.ipynb

Objective

When conv2_main (phase03's most necessary component) is pushed toward the promoter direction, how does that perturbation actually reach the final readout -- one concentrated pathway, or several redundant ones?

Method

A promoter direction estimated at conv2_main is injected into 80 non-promoter recipients. Stage-wise propagation is traced using phase01's per-depth probes. Individual and joint mediation test whether restoring each of 5 downstream components (conv4_main, conv7_main, xformer1_fc2, deconv1_main, deconv4_main) to its clean value blocks the propagated effect at the final readout (deconv7).

Key results

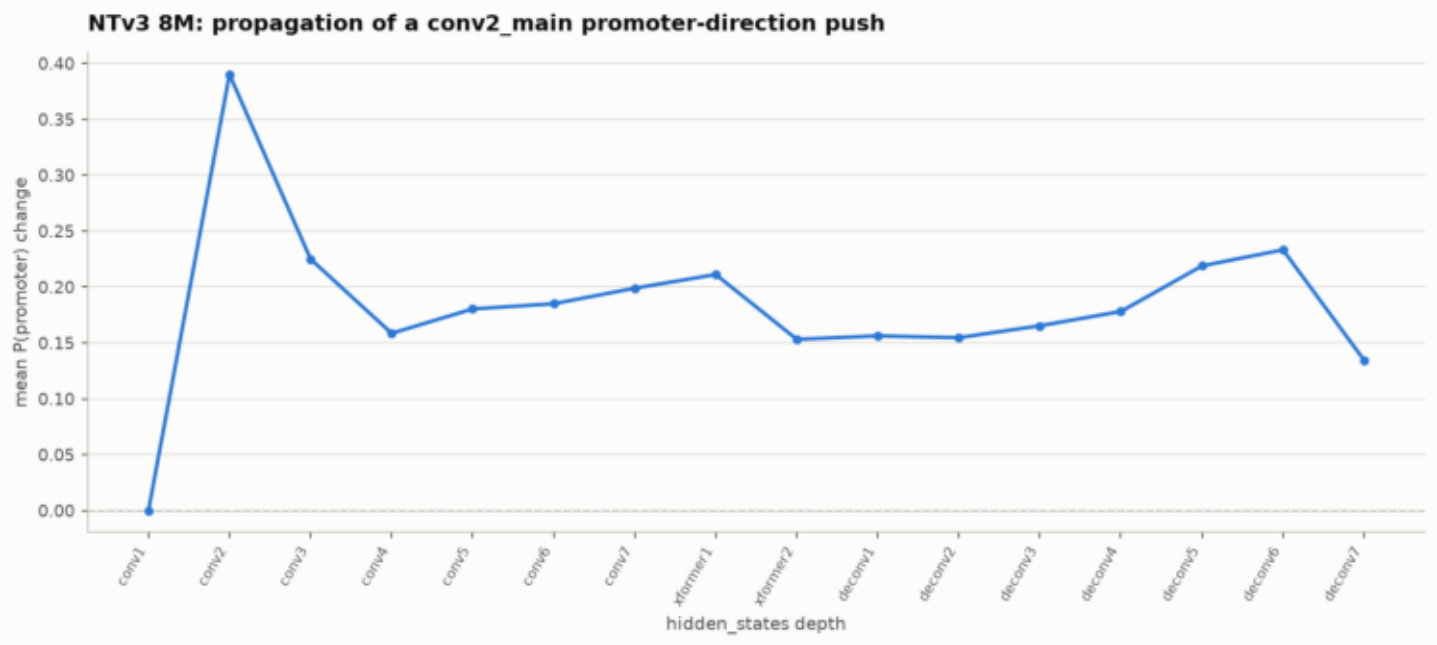
Total effect at final readout: +0.1346

Individual mediation (mean blocked effect, fraction of total):

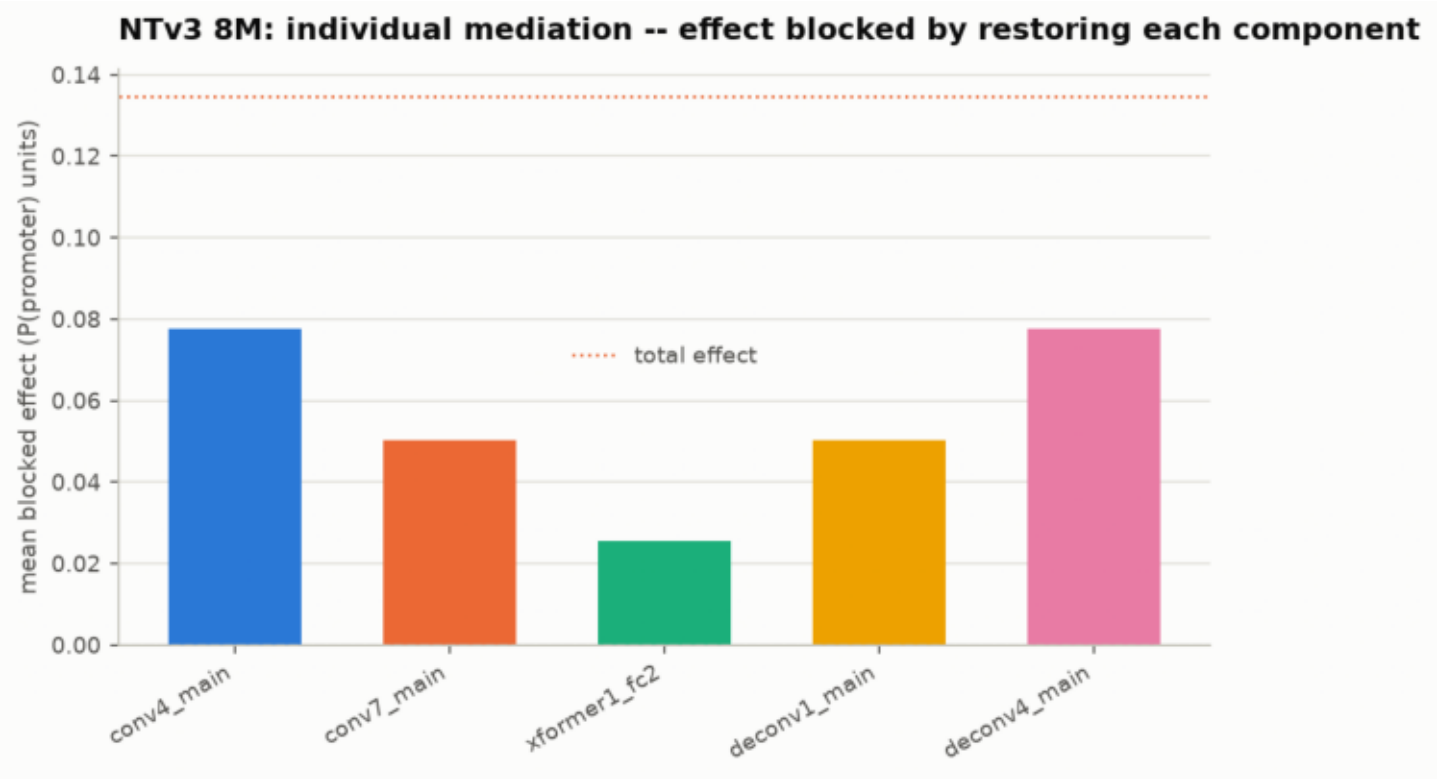
conv4_main	+0.0776	(+58% of total)
conv7_main	+0.0503	(+37% of total)
xformer1_fc2	+0.0256	(+19% of total)
deconv1_main	+0.0503	(+37% of total)
deconv4_main	+0.0776	(+58% of total)

Interpretation

conv4_main individually blocks the majority (58%) of the propagated effect at the final readout. Caveat: every joint-restoration condition tested here produced exactly the same blocked effect as restoring conv4_main alone -- consistent with the downstream-overwrite screening property described in this module's docstring (all five candidates sit on one serial chain), not necessarily genuine redundancy in the deeper sense. This scan cannot distinguish the two without a mediation method that preserves upstream contributions through skip connections.



phase09 stage propagation



phase09 individual mediation

Notes

Architecture note

NTv3 8M is a U-Net: a 7-stage convolutional downsampling tower feeds a 2-layer Transformer bottleneck, upsampled back to full resolution by a matched 7-stage deconv tower. At this study's 70nt promoter length (padded to 128nt), the Transformer bottleneck collapses to a single token, so phases 03-09 examine conv/deconv components and skip connections rather than attention heads.

Reproducing this report

Every figure and number above was produced by running experiments/phase01_representation_baseline.p through phase09_pathway_mediation_validation.py against the real InstaDeepAI/NTv3_8M_pre checkpoint (gated on Hugging Face -- see README.md for access setup), then this script (ntv3-interpretability/build_report.py).

Caveats worth carrying forward

Phase03's necessity/mediation results are only causally meaningful at the readout depth used (the final one); phase07's unsupervised subspace is less predictive than phase06's supervised direction; phase09's joint-mediation numbers can't fully separate genuine redundancy from a hook-mechanics screening effect on a serial computational chain -- see each phase's own module docstring for the full reasoning.